3244-1910-0020-4 - Machine Learning with a Heart

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Abstract

Our project aims to predict if a particular patient has a heart disease given their health information. It also seeks to reveal insights on the significant factors that affect the chances of getting a heart disease.

Our project can be used to give a rough estimate to individuals who wish to check if they are at risk of heart disease. They can then take the necessary precautions to address the issues they might have in their lifestyle to reduce that risk. This will help to educate the public on heart disease and the factors that contribute to it, and can provide guidelines to them on how to live a healthier lifestyle.

We will be utilizing various Machine Learning algorithms learned in class such as Random Forest, Decision Trees, Naive Bayes etc, and evaluate the performances based on Accuracy and Area Under Curve.

Introduction

Motivation & Importance of the Project

The amount of data generated today is so extensive that 90% of all data has merely been generated in the last 2 years alone [1]. The advent of data has pushed the effectiveness of machine learning techniques to its peak.

This has immensely helped the healthcare field, where some Machine Learning classification models perform better than trained clinicians [2].

However, with some specific diseases, the sample sizes may be too small, which would reduce the effectiveness of extrapolating the model to the general population.

Heart disease was chosen as it is the number one cause of deaths worldwide, causing 31% of all global deaths [3], and the detection of such a disease is a complex task, which requires numerous amount of effort and time from the medical personnel.

By helping to increase heart disease detection by leveraging on data, we may enhance the efficiency of doctors, thus allowing them to work on aiding the patient immediately.

Furthermore, increased heart disease detection may enable medical authorities to provide precautionary measures or treatment as soon as possible, thus increasing their lifespans.

Related Work

There have been numerous works by many researchers in attempting to predict heart disease using data-mining techniques such as clustering or optimization algorithms, such as MAPO (Modified Plant Optimization).

The paper [4] proposed by Prerna Sharna, Krishna Choudhary, Kshitij Gupta used MAPO to detect heart rate using fingertip video and subsequently, Logistic Regression, Naive Bayes and XGB (Extreme Gradient Boosting) to calculate the presence of heart disease.

Another paper [5] uses Multi-Layered Perceptron (MLPs) to predict heart disease, using data collected from Fitbits and other health-related applications.

A similar paper [6] analyzed the same dataset as this report. The paper obtained a classification accuracy of 85% using logistic regression. To further enhance on this paper, we have decided to use more complex models, such as XGBoost. We will also use this paper as a benchmark and compare their results with ours.

Method

The first step to any machine learning process is data cleaning and exploration. For data cleaning, rows with missing values, improper data format as well as outliers were removed. Numerical features and Categorical features were then separated for data exploration. Bivariate analysis was then performed for each variable, conditioned upon have HD or no HD.

Continuous Data Exploration

For continuous variables, we made use of quantile-plots, box-plots (Figure 1) and statistical measures such as standard deviation and median (Figure 2) to identify the feature values at different quantiles.

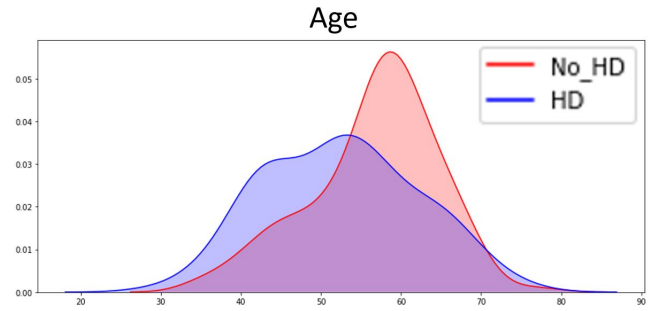


Fig 1. Distribution Plot for Ages (Separated by HD Status)

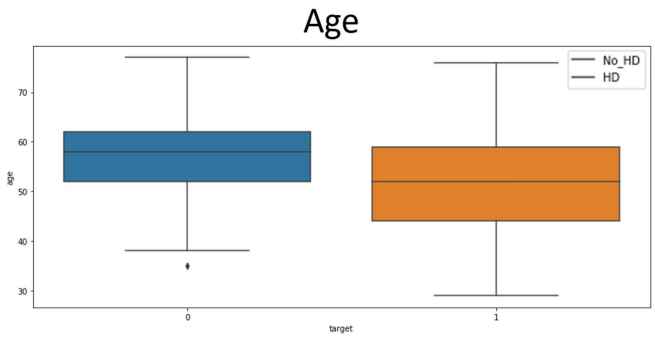


Fig 2. Box Plot of Age conditioned on having HD or no HD

Categorical Data Cleaning

For categorical variables, we plotted the normalized and unnormalized probability density plots using histograms to identify the proportions of unique values within each variable (Figure 3).

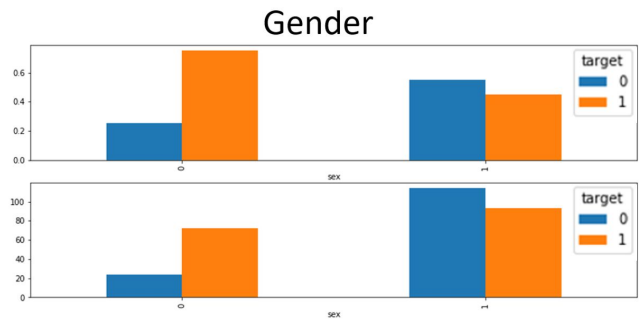


Fig 3. Proportions (top) and Normalized Proportions of Labels based on Gender

To assess if we have unbalanced label data, the proportions were plotted (Figure 4). Our data consists of 54% patients with HD and 46% patients without HD. As the proportion of labels are relatively equal, we do not need to perform any over-sampling methods to re-balance our labels

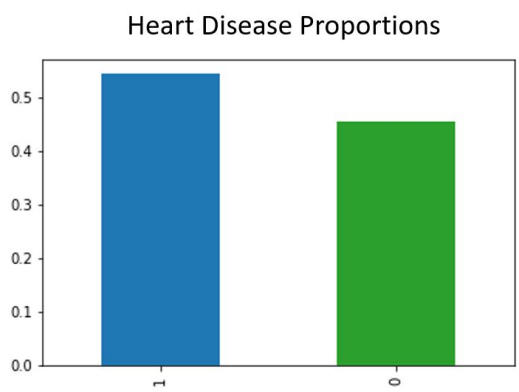


Fig 4 Proportions of Labels

Feature Significance and Selection

For feature selection, statistical methods such as Pearson's chi-squared test was used to remove independent features for categorical variables. Conversely, the Mann-Whitney U test was used for continuous variables. Both of these tests were carried out at the 5% significance level. The test values are in Appendix B.

Our final dataset consists of 304 rows and 14 columns of features.

Encoding Categorical Variables

For categorical variables, we performed **one-hot-encoding** such that each categorical feature has its own binary featured columns.

Despite tree-based methods not requiring such data preprocessing, our team decided to perform it anyway because other models such as Logistic Regression, are unable to distinguish the difference between continuous and multi-leveled categorical variables.

Having sparse data due to one-hot-encoding was not a huge issue in our dataset as we do not have a lot of features. There are also other methods of categorical variable encoding such as mean encoding. However, we did not carry it out as mean encoding may result in data leakage, thus jeopardising our model generalisation [7].

Train-test Split

The dataset was split into training (75%) and testing (25%). We performed 5-fold cross validation on the training set to obtain the best hyper-parameters for the various models. 5-fold cross validation was then used to assess the predictability of the models according to AUC (Area Under Curve) and accuracy.

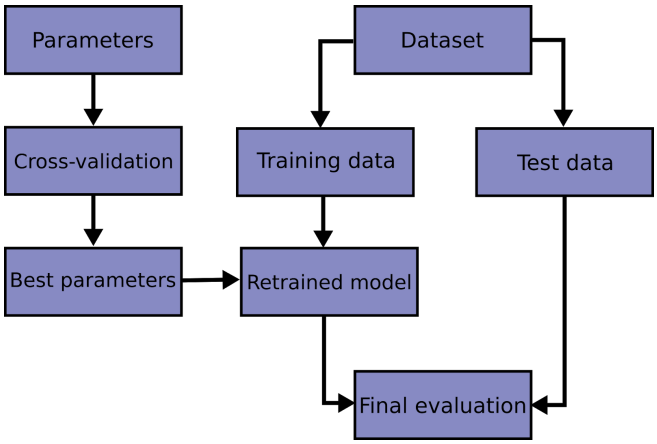


Fig 5 Cross Validation Workflow [8]

For our models, we made sure to include both complex models, such as XGBoost, and simple models, such as Logistic Regression. This allows us to compare the effectiveness of simple against complex models. In total, 8 models have been used for evaluation and the results are shown in Figure 10.

Analysis of Top-Performing Models

After training our models, we selected the top-3 best performing models according to our test AUC scores. We then plotted AUC based on one-fold of data to assess how well our model is able to handle the trade off between True Positives (TPs) and False Positives (FPs).

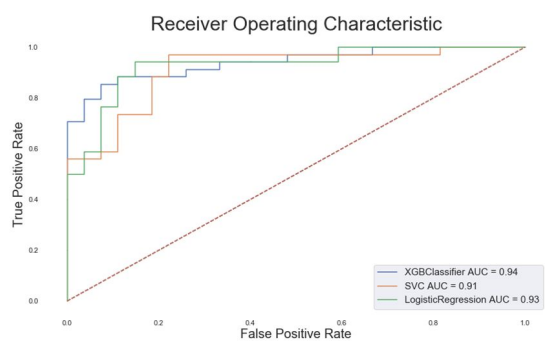


Fig. 6. ROC Plot for Best-3 Models

Further discussion on why and how we use AUC to evaluate our model is elaborated below.

Evaluation

Oftentimes, many of us are obsessed with knowing the accuracy of models. However, accuracy is inferior to probability-based evaluation for fields such as Medicine as predicting an 80% probability of a person having HD does not mean that a person **will have** HD. What medical authorities are concerned with is the relative probability of the patient having heart disease versus not having heart disease.

To evaluate this, we simply need to check if our model is consistently predicting higher probabilities of heart disease in patients who do have it, and lower probabilities of heart disease in those that do not. Then, we would be able to correctly classify most patients as having heart disease or not by referring to a threshold.

Accuracy presumes equal importance of the 4 different outcomes of prediction: (1) True Positive (TP), (2) False Positive (FP), (3) False Negative (FN) and (4) True Negative (TN).

This is a pitfall for medical prediction as medical authorities place a **larger emphasis** on **TPs** and **FNs**. Having a TP indicates that authorities are able to convey treatment to patients that have a high chance of HD whereas a FN will mean that authorities had denied a patient vital medical aid.

In summary, a TP is a correct diagnosis that can save a life and a FN is a misdiagnosis which may have cost a life.

To emphasis greater importance on TP and FN, the AUC-ROC (Area Under the Receiver Operating Characteristics) curve shall be used.

How to make use of AUC

AUC-ROC is a performance measurement for classification problem set at different threshold settings [9]. Generally the larger the AUC, the better the model as this indicates a better trade-off between TPs and FPs.

The ROC curve is plotted with True Positive Rate (TPR) against False Positive Rate (FPR). Since False Negative Rate = (1 - True Positive Rate), a higher True Positive Rate will indicate a smaller False Negative Rate. Both rates are important prediction performance indicators for medical authorities.

Figure 7 gives an intuition as to why a larger AUC is desirable. For a fixed FPR, the 3 models have 3 different TPRs. The model with a larger area under the curve (marked in Figure 7) has the largest TPR, and thus the smallest FNR.

It is observed from Figure 7 that since the Linear Discriminant Analysis (LDA) model has the largest TPR, it also has the smallest FNR. Hence, this is why LDA is the best-performing model, **when FPR = 0.1.**

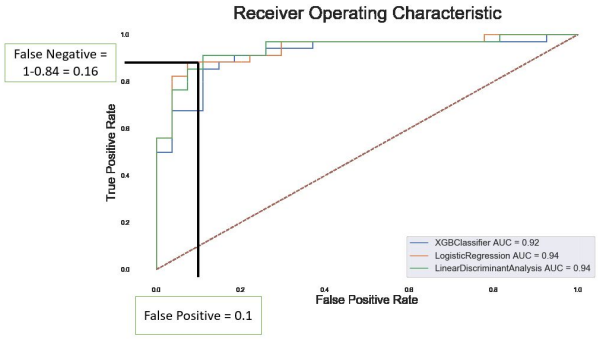


Fig. 7. ROC Plot for Best-3 Models, with Fixed FPR at 0.1

As the ROC-curve plots the FPR by changing the threshold for prediction, this highlights the importance of defining thresholds for model selection. There is no single model that is the best for all thresholds. Figure 8 indicates that Logistic Regression is the best-performing model, when **FPR = 0.05.**

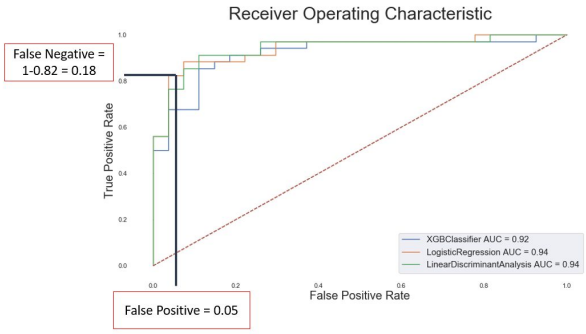


Fig. 8. ROC Plot for Best-3 Models, with Fixed FPR at 0.05

An interesting contextual fact to note is that although both TPs and FNs are the main performance indicators of the prediction models, the other indicators should not be neglected. False Positives incur an unnecessary medical and emotional cost for both patients and medical bodies whereas True Negatives measure the actual patients who do not enjoy unnecessary resources.

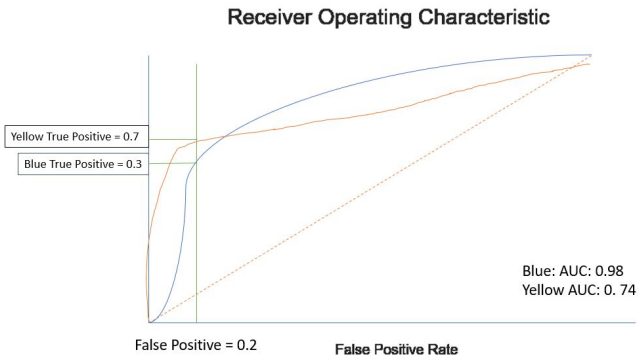
Hence, False Positives and True Negatives must be considered and balanced with True Positives and False Negatives using the AUR-ROC curve.

Suppose that True Positives and False Negatives are the only necessary evaluation metrics required. a sufficient argument will be to predict that all customers have a high chance of having HD. In this case, our True Positives will be perfectly 1 and False Negatives will be 0. However, from our AUC, FPR will increase along with the proportion of misdiagnosis of our patients. By misdiagnosing more patients, this reflects poorly of the hospital standards. Hence, we should not solely consider TP and FN as a performance measure.

To conclude, while AUC is able to indicate the effectiveness of a model in a trade-off between True Positives and False Positives, it

is even more important to decide what threshold level to set. This is because a model with a lower AUC score may have a higher TPR at a given FPR than a model with a higher AUC score (Figure 9).

To fully utilize machine learning models, medical authorities must decide the threshold wisely, i.e. how much False Positive Rate they are willing to tolerate and their desired True Positive Rate and False Negative Rates. With a specified threshold, they can then decide on an appropriate model to deploy.



Fg. 9. ROC Plot showing that although AUC is high, at a given FPR it is possible that the better model is the one with lower AUC

Further Analysis (Feature Engineering)

After our initial modelling, we re-examined the data and found that there are only 4 instances that have "restecg" = 2, with the other 2 values being equally represented.

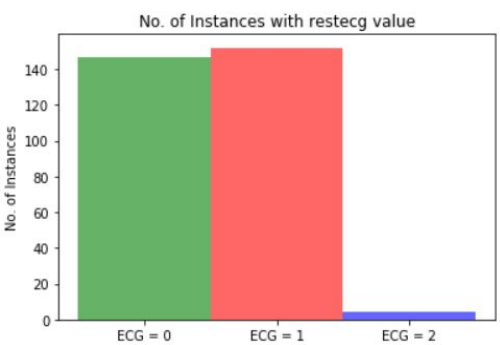


Fig 10. Distribution of data split by ECG levels

Upon further research, we found that for both categories of "restecg" that signify abnormal readings (represented by values 1 and 2), both these values are found to increase the risks of heart disease.

Thus, due to the very few samples of "restecg" = 2 that we had, we replaced those values with 1, grouping the two abnormal types of readings. The new "restecg" column contains only 2 unique values of "restecg", 0 indicating a normal reading and 1 indicating an abnormal reading.

Although the newly modified data got a higher accuracy for the training set, it fared worse against the test. Thus, we decided to exclude it from the final model.

Dataset

Data was obtained from the 4 different medical facilities; Cleveland Clinical Foundation (Ohio, USA), Hungarian Institute of Cardiology (Budapect), V.A. Medical Center (California, USA) and University Hospital Zurich (Switzerland).

Data contained 303 instances, with 13 attributes (14 including labels). Attributes are a mix of categorical, integer and real valued variables.

A full description of the dataset is in the poster, as well as in Appendix A.

Macroscopic

The following figures shows the Mean AUC from the cross-validation of our training data, and the AUC from the 25% of our testing data.

Models	Accuracy	Accuracy_Std	AUC	AUC_Std
LinearDiscriminantAnalysis	0.837796	0.0649618	0.908222	0.030928
LogisticRegression	0.841217	0.0507396	0.903005	0.0365893
XGBClassifier	0.81823	0.0470777	0.888211	0.0455219
GaussianNB	0.775071	0.0697621	0.872689	0.0336918
RandomForestClassifier	0.762086	0.0379987	0.841169	0.0414167
DecisionTreeClassifier	0.735943	0.0452171	0.721524	0.0567116
KNeighborsClassifier	0.640448	0.0177057	0.666766	0.023525
SVC	0.567595	0.018087	0.608496	0.0497695

Fig. 11. 5-Fold Cross Validation Results of all Models

Models	AUC
LinearDiscriminantAnalysis	0.94
LogisticRegression	0.94
XGBClassifier	0.92

Fig 12. AUC Validation of Top-3 Model on Testset

Contrary to our initial hypothesis that the state-of-the-art XGBoost and complex models such as RandomForest will be the best models, Logistic Regression actually outperformed the other models in terms of **accuracy**, and Linear Discriminant Analysis outperformed the other models in terms of **AUC**.

This phenomenon can be explained by the figure below depicting the comparison between simple and complex methods with respect to the number of data samples. We can clearly see that with a **small data set**, the test performance of simple models are better than complex models.

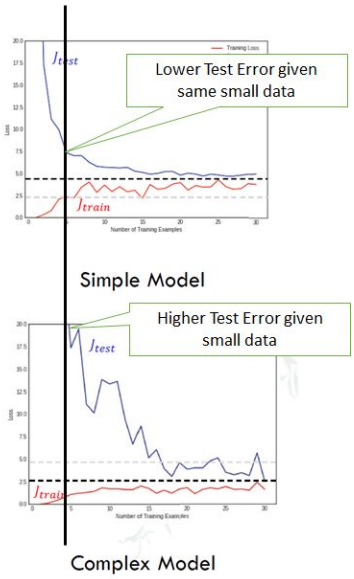


Fig 13. Simple Models Outperforming Complex Models with Small Datasets

Overall, we decided that LDA is our best model because although both LDA and Logistic Regression have the same AUC score on the testset (Figure 12), the mean AUC score of LDA from our cross-validation is higher (Figure 11).

With that being said, one should keep in mind about what was being mentioned about thresholds in the evaluation part of this report when selecting models.

Other Evaluation Metrics

From the evaluation section, we have understood the importance of threshold selection for True Positive Rates and False Positive Rates.

Our team is aware of other metrics such as Precision, Recall or F1-Score; which measures the balance between precision and recall. However, our team decided not to use these performance measures as they are dependent on the threshold that is being decided for deployment.

Without a specified threshold, it is difficult to determine the best model based on these metrics.

Discussion

From what we have learnt about decision trees, decision trees are good at identifying which are the more important variables to classify the data. Thus, to look into the most important factors that affect the chances of getting heart disease, we decided to look at the Feature Importance Plots generated from XGBoost. Since XGBoost is a Decision-Tree based ensemble method that utilises information theory [14] for the selection of variables, it will give us some insights on the extent to which factors affect the chances of getting heart disease.

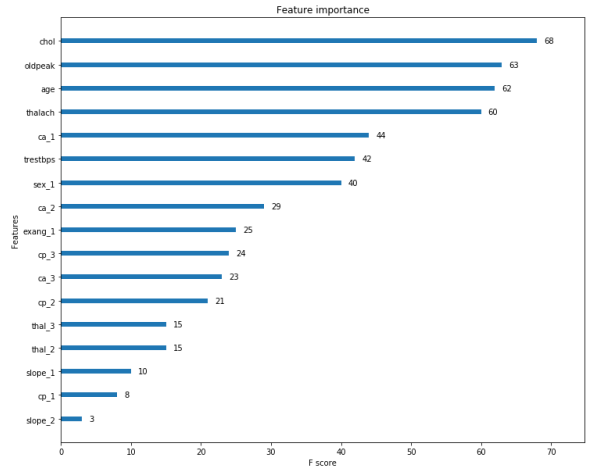


Fig 14. Feature Importance Plot of Variables

With our best model (LDA), we conducted Partial Dependence Plots (PDP) for each non-categorical variable. PDPs indicate how each variable will affect the model's predictions [13].

With the plots, we can observe which variables are the most significant in predicting HD. The full plots are in Appendix C.

To select which variables are more important to discuss, we selected the features that were indicated as the 3 most important features in the Feature Importance Plot above.

Here are the PDPs of the 3 selected variables with the highest feature importance.

Partial Dependence Plots

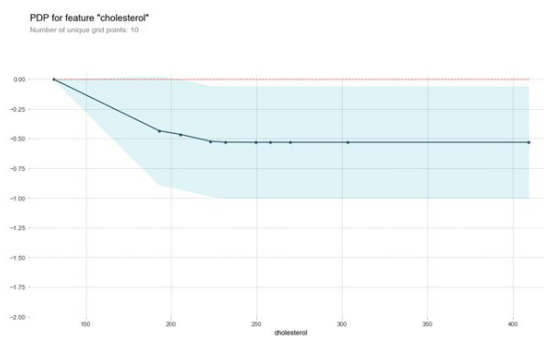


Fig 15. PDP for Variable: Serum Cholesterol

From Figure 15, it is observed that as serum cholesterol levels increase, the lower the likelihood of heart disease.

This contradicts with common intuition. However, further research indicates that serum cholesterol can be divided into 2 categories; high-density lipoprotein (HDL) and low-density lipoprotein (LDL).

LDL cholesterol can build up in a person's arteries, clogging them and reducing blood flow thus causing heart disease. Whereas HDL has the opposite effect whereby the higher the amount of HDL is, the lower the chance to have a heart disease [11].

Since the serum cholesterol is calculated by the total amount of cholesterol present, it is not a good indicator that we can use to predict heart disease. Using a ratio of HDL to LDL would be a better measurement in this case.

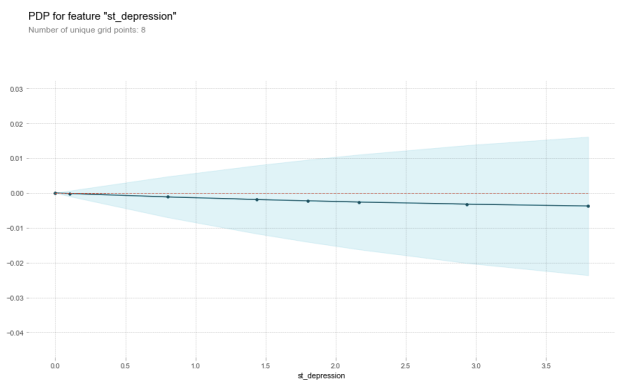


Fig 16. PDP for Variable: ST depression induced by exercise relative to rest

From Figure 16, it is observed that as “oldpeak” / “st_depression” (an exercise induced indicator) increases, the probability of heart disease decreases.

However, this trend is against what research suggests [15]. Perhaps this is due to our relatively small dataset.

Furthermore, as the change is very close to 0, with more data, we may see the correct trend being exhibited in the PDP ie an upward sloping curve.

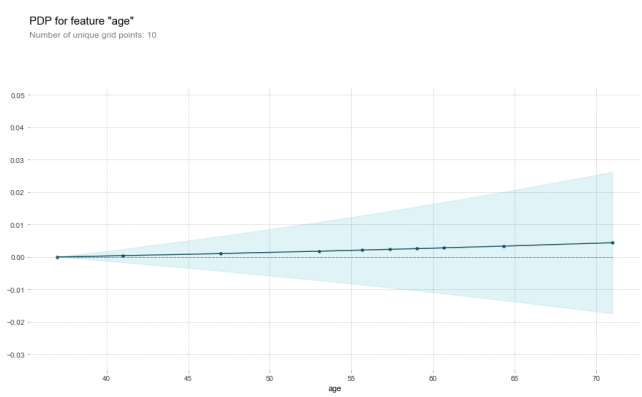


Fig 17. PDP for Variable: Age

From Figure 17, it is observed that as “age” increases, the probability increases. This agrees with research [16], where the older an individual is, the higher the likelihood of heart disease.

Conclusion

With such insights, preliminary guidelines or cautionary measures can be provided to heart disease centres to advise people to keep a lookout for these significant factors with respect to their own health.
Health centres can propagate the message possibly through health booklets or posters.

Learning from the small amount of data that we used, heart disease centers should also ensure that they have sufficient quality data for model to train on. Assuming that If heart disease centers were to utilize a machine learning model for prediction, it is also extremely important to be able to decide on the threshold for classification

Shortcomings

Even though we have obtained a relatively high accuracy and AUC, we must be careful when generalizing to the public due to the low sample size.

Furthermore, as the dataset we used had already been filtered by the competition organizers, we had originally started with only 120 rows of data and our dataset is extremely small compared to other teams. Using the advice from our Interim consultation, we tried to source for the original dataset that was being used, but we were only able to increase our data from 120 rows to 300 rows (by finding the raw data repository). Although we have more than doubled our initial number of data, eventually at the end of our project, we realized that it is still insufficient because our simple models are performing better than complex models.

We understand that some teams have attained data through Web Scraping. However, as patient information is highly confidential, we were unable to find websites that publicly made these data available.

Furthermore, other publicly available datasets, such as [12], have datasets which are much larger (70, 000 instances). However, the other datasets were collected to model different problems. The dataset from [12] was used to predict cardiovascular disease, not heart disease.

Hence, it would be better to not absorb different datasets to avoid data contamination.

Even though the aim of our project was to understand more about heart disease, we were unable to fully comprehend the results that was shown in our project. The effect of Serum Cholesterol was contradictory to general knowledge and there were many conflicting research conclusions as to whether or not cholesterol has an association with HD. As of any machine learning problem, the best way to really understand these variables will be to seek medical domain experts.

Future Work

To improve on our current model, more data can be collected to augment the dataset. This will increase the accuracy of the model and improve the generalizability of the model.

With more data, more sophisticated Machine Learning techniques can be used, such as Deep Learning and Neural Networks, which might be able to provide more insights on heart disease and its contributing factors.

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Appendix A: Variable Analysis

Variable	Type	Variable Values	Variable Explanation (How does it affect heart disease)
AGE	Numeric	Age of the individual, in years.	The older, the higher the risk of heart disease
SEX	Categorical	Gender of the individual(1 = Male, 0 = Female)	Different factors(eg. diabetes) affect the different genders differently in terms of risk of getting heart disease
TRESTBPS	Numeric	Resting blood pressure of the individual on admission to the hospital.	The higher the resting blood pressure, the higher the likelihood of heart disease
RESTECG	Categorical	Electrical heart activity of the individual; detects ST-T wave abnormality & left ventricular hypertrophy.	ST-T wave abnormality suggests an increased risk of heart disease. Left ventricular hypertrophy puts more stress on the left ventricle, which over time, might cause it to fail, and suggests an increased risk of heart disease.
THALACH	Numeric	Max heart rate achieved.	Higher maximum heart rate, lower likelihood of heart disease.
CA	Categorical	No. of major blood vessels; colored by fluoroscopy. (Ranges from 0 to 3)	More coloring suggests a more certain amount of blockage in that specific blood vessel which means a higher likelihood of heart disease
THALl	Categorical	Indicates how well blood flows into the heart. (3 = normal; 6 = fixed defect; 7 = reversible defect)	Fixed defect means scarred myocardium from prior infarct will not take up the tracer and thus a higher likelihood of heart disease. Reversible defect means coronary stenosis when a myocardial segment takes up the tracer at rest but not during cardiac stress. This is a good indicator of heart disease.
CP	Categorical	Chest pain type. (1 = Typical Angina, 2 = Atypical Angina, 3 = Non-anginal Pain 4 = Asymptomatic)	*Typical angina* occurs usually when the heart does not get enough blood as it needs. This occurs due to the presence of heart disease. *Atypical chest pain* is a term used to describe discomfort or pain centered in the chest that is not cardiac pain nor heart (disease) related. Chest pain is very likely *nonanginal* if its duration is over 30 minutes or less than 5 seconds. It is also not related to heart disease. *Asymptomatic* refers to heart disease/attack that does not display any symptoms.
CHOL	Numeric	Measurement of high/low-density lipoprotein cholesterol.	LDL cholesterol can build up in a person's arteries, clogging them and reducing blood flow thus causing heart disease.
SLOPE	Categorical	How the ST segment behaves during exercise. (1 = Upsloping, 2 = Flat, 3 = Downsloping)	Upsloping is the normal ST-segment during exercise. A flat Horizontal ST-segment depression of ≥ 2 mm indicates a high probability of heart disease Downsloping indicates a high probability of heart disease.
OLDPEAK	Numeric	ST Depression Induced By Exercise Relative To Rest.	ST-segment depression induced by exercise is a very strong predictor of sudden cardiac death in men with any conventional risk factor but no previously diagnosed heart disease.
FBS	Categorical	Fasting blood sugar level. (1 = > 120mg/dl, 0 = < 120mg/dl)	People with a fasting blood sugar level of 100-125 mg/dl had an adjusted nearly 300% increase higher risk of having coronary heart disease than people with a level below 79 mg/dl.
EXANG	Categorical	(1 = Yes, 0 = No)	Angina that is caused by physical exertion.

Table 1. Variable Analysis

Appendix B: Feature Selection Tests

Categorical Features	Test Statistic
sex	**Chi-squared Statistic:** 23.914383914761988 **p-value:** 6.412944979367393e-06
cp	**Chi-squared Statistic:** 81.68642755194445 **p-value:** 1.3343043373050064e-17
fbs	**Chi-squared Statistic:** 0.23832909479016695 **p-value:** 0.887661726319414
restecg	**Chi-squared Statistic:** 10.023091785081 **p-value:** 0.018370863234825497
exang	**Chi-squared Statistic:** 57.7992952073631 **p-value:** 2.8121761247102573e-13
slope	**Chi-squared Statistic:** 47.506896756030244 **p-value:** 2.7114148545147224e-10
ca	**Chi-squared Statistic:** 72.92217203269523 **p-value:** 1.01008763718801e-15
thal	**Chi-squared Statistic:** 85.30373951466149 **p-value:** 2.233350721012921e-18

Fig 18. Pearson chi-squared Test Values

Numerical Features	Test Statistic
Age	**Statistics**=8240.500, **p**=0.000
Trestbps	**Statistics**=9784.500, **p**=0.017
Chol	**Statistics**=9789.500, **p**=0.018
thalach	**Statistics**=5732.000, **p**=0.000
oldpeak	**Statistics**=5922.000, **p**=0.000

Fig 19. Mann-Whitney U Test Values

Appendix C: Partial Dependence Plots

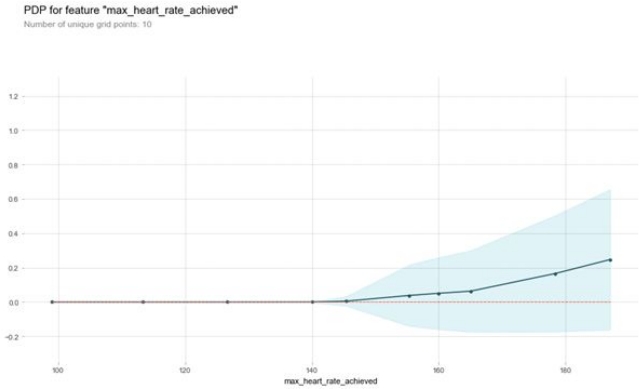

Fig 20. Partial Dependency Plot for Max_Heart_Rate

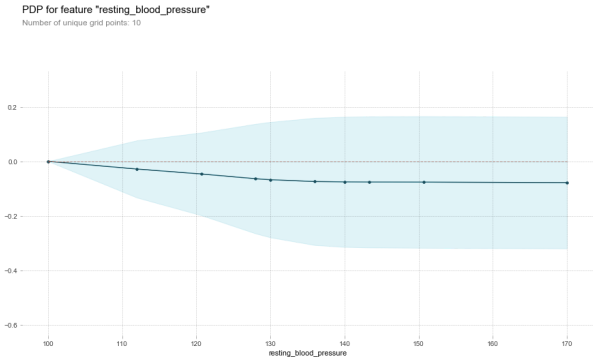

Fig 21. Partial Dependency Plot for Resting_Blood_Pressure

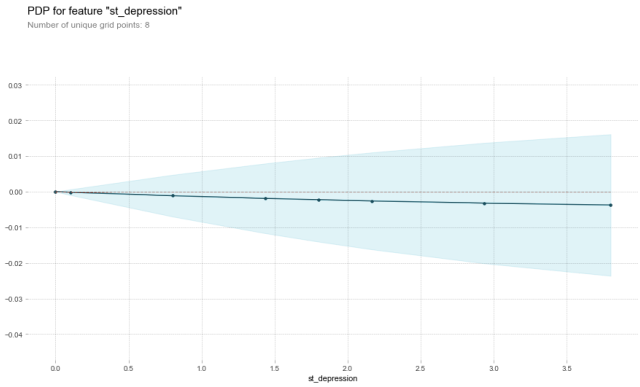

Fig 22. Partial Dependency Plot for St_Depression